

Supplementary Information:

IRSpectra-Bench: protocol, tables, cases, and reproducibility

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Companion to the ICLR 2027 submission

Molecular Structure Elucidation with Frontier Models:

A Benchmark on Literature-Reported IR, ^1H , and ^{13}C NMR Peak Lists

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This supplement records the blind protocol, scoring identities, scale tables, worked cases, solver-facing contracts, and reproducibility notes that exceed the ICLR main-text limit. Headline generation on the frozen $n=500$ roster is 227/500 (45.4%) top-1 and 249/500 (49.8%) recall; self-ranking precision | recall is 227/249 (91.2%). Independent forward verification of the same pools is Figure S2 and §8. Confidence intervals are not reported for $n=500$. The $n=519$ pooled print (243/519) is not the headline. Unprefixed figure and listing numbers refer to the main manuscript.

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1 Cross-reference to the main text

Table S1: The map shows main-text objects and their counterparts in this supplement.

main label	object	role here
Fig. fig1	Fig. 1 (fig1_lead_overview)	Protocol and $n=500$ generation: 249/500 recall $\times$ 227/249 self-rank = 45.4% top-1; wall 227/22/251. Reprinted as Fig. S1.
Fig. fig-chemspace	moved here	Literature-cohort chemistry (median MW 306; median 2 rings). Reprinted as Fig. S3.
Listing lst:payload	Listing 1 (live lstlisting + PubChem 3D)	IRexp JSON+molecule (ILJNJNKEOAREX; CID 57398578; C ₁₂ H ₁₁ BrN ₄). SMILES and InChIKey are resource fields, not solver payload. Reprinted below.
Fig. fig-wall	appendix (fig_wall_diagnostic)	$n=500$ forward-verify wall 204/45/251. Reprinted as Fig. S2.
Fig. fig-robustness	Fig. 2	Formula-only, recency, and four-vendor controls (not reprinted).
Fig. fig-mechanism	Fig. 3	Locked v3-R25 verify-save. Reused in §9.
Fig. fig3-method	moved here	Inference ladder on the 60-compound arm (23% $\rightarrow$ 27% $\rightarrow$ 30%). Reprinted as Fig. S4.
App. cases	Table tab:cases (kept in main appendix)	Same integers as §9.
validate-clean extras	main headline one-liner; tables here	Scaffold / Tanimoto / stereo / size bins / corpus-reweight (§5). Not reprinted as appendix prose.
lit. decomp. table	moved here (§8 onward / literature)	Full top-1 / top- k rows; main keeps the 68/70/83% collapse.

Main-text Eqs. (1)–(6) (factorisation, InChIKey-14, top-1, recall@ k , prec./recall, ^{13}C chamfer) are restated in §4 with the same meaning. The pooled estimator (former main-text Eq. (7)) lives only here.

2 Reprinted figures

Figures S1 and S2 reprint the main-text protocol plate and the $n=500$ wall so the supplement can be read alone. Listing 1 is reprinted with the same live lstlisting+framed layout (PubChem 3D, CID 57398578). Cohort chemistry and the 60-compound inference ladder live only here so the ICLR body can keep the narrative.



Figure S1: The reprint shows main-text Figure 1: generate 249/500 (49.8%) recall in top 3; select 227/249 (91.2%) top-1 | recalled ($49.8\% \times 91.2\% = 45.4\% \text{ top-1}$); wall 227 exact top-1 / 22 in-pool not top-1 / 251 never proposed.

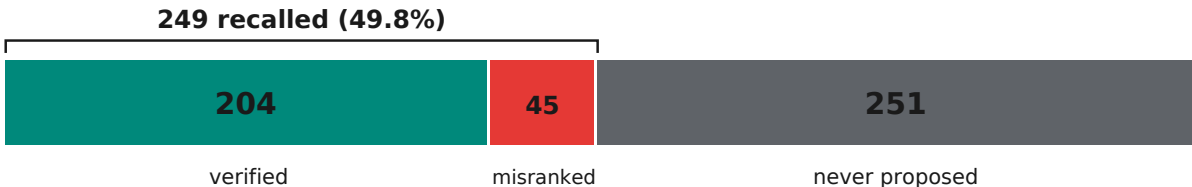


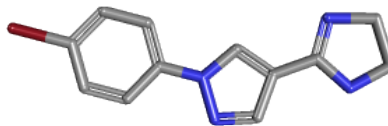
Figure S2: The reprint shows the $n=500$ forward-verify wall: 204 verified / 45 misranked / 251 never proposed; generation remains 227/500 top-1 and 249/500 recall.

3 Blind protocol

3.1 Solver contract

Each IRSpectra-Bench item is a four-field JSON object. The solver sees only

```
{
  "id": "ILJNJNKEOAREX-UHFFFAOYSA-N",
  "formula": "C12H11BrN4",
  "ir_bands_cm-1": [3144.0, 3110.0, 3066.0, 2931.0,
    2863.0, 1632.0, 1589.0, 1565.0, 1485.0],
  "ir_source": "experimental",
  "source_doi": "PMC:12566713",
  "pmcid": "PMC12566713",
  "h_nmr": "8.83 (s, 1H), 8.04 (s, 1H), 7.81 (d, J = 8.8 Hz, 2H),
    7.72 (d, J = 8.8 Hz, 2H), 3.56 (s, 4H)",
  "c_nmr": "157.5, 140.3, 138.4, 132.4, 127.5, 120.4, 119.0, 116.4, 49.1",
  "smiles": "BrC1ccc(-n2cc(C3=NCCN3)cn2)cc1",
  "inchikey": "ILJNJNKEOAREX-UHFFFAOYSA-N",
  "has_structure": true,
  "license": "CC-BY",
  "license_pool": "commercial"
}
```



Listing 1: The reprint shows main-text Listing 1, an IReXP record (commercial DoR; ILJNJNKEOAREX; PubChem CID 57398578): SMILES and InChIKey are resource fields, not solver payload.

{qid, formula, ir_bands_cm-1, h_nmr, c_nmr}.

No name, SMILES, InChIKey, scaffold, atom-type hint, retrieved spectrum, licence stamp, or PMC id enters the prompt (main-text Listing 1, adapted from one IReXP record). Gold constitution is scored offline by the InChIKey-14 indicator (§4).

Difficulty is declared a priori by RDKit ring analysis: simple iff ≤ 2 rings, no fused/spiro/bridge-head system, ≤ 22 heavy atoms; else complex. The $n=500$ headline is near-50/50 (248/252); the eligible IReXP corpus is 17.5% simple / 82.5% complex. Corpus-reweighted extras on a validate-clean subset are in §5 and are not $n=500$ figures.

3.2 Pre-registration

The code release records the frozen roster and the expand-500 campaign.

- $n=500$ roster: frozen qid lists in the code release. Sampler seeds for early draws were not captured; the released roster is fixed and reusable but not exactly re-drawable.
- expand-500 campaign: 115/115 simple/complex; seed 2026500; 0 collisions with any prior round. Six ^{13}C -overread flags known before any solve: R26, R31, R102, R105, R107, R138.

3.3 Stopping and pooling

- The $n=500$ roster is a pre-specified frozen qid list, not an outcome-selected subset.
- Gold keys for the expand-500 arm were used only for local scoring and are not in the review copy.
- Official expand-500 chamfer is 103/230 verify vs 129/230 self (§8); confidence intervals were not computed on that arm.

3.4 Allowed tools

RDKit formula / parse check only. No web search. No retrieved spectrum. No thinking tier on the core protocol. A thinking-tier consumer arm is labelled as such in the score notes and is not interchangeable with the no-thinking protocol.

A deposit is JSONL of {qid, candidates: [SMILES,...]} under a stated budget. Headline budget: ≤ 3 ranked SMILES per qid. Self-ranking and forward-verified re-ranking are reported separately.

4 Scoring identities

These identities match main-text Eqs. (1)–(7).

A candidate $\hat{y}$ matches gold y^* at constitution iff

$$s(\hat{y}, y^*) = \mathbf{1}[\text{IK}_{14}(\hat{y}) = \text{IK}_{14}(y^*)]. \quad (\text{S1})$$

IK_{14} is the first 14 characters of the RDKit InChIKey (connectivity layer). Full stereo-sensitive InChIKey is reported only on validate-clean $n=295$ (§5).

With ranked pool $\mathcal{C}_i = (c_{i1}, \dots, c_{ik})$, $k \leq 3$,

$$\text{top-1} = \frac{1}{n} \sum_i s(c_{i1}, y_i^*), \quad (\text{S2})$$

$$\text{recall}@k = \frac{1}{n} \sum_i \mathbf{1}[\exists j \leq k : s(c_{ij}, y_i^*) = 1], \quad (\text{S3})$$

$$\text{prec.}|\text{recall} = \frac{\sum_i s(c_{i1}, y_i^*)}{\sum_i \mathbf{1}[y_i^* \in \mathcal{C}_i]}. \quad (\text{S4})$$

These are the three reportable numbers. They factorise as

$$\text{top-1} = (\text{recall}@k) \times (\text{prec.}|\text{recall}) \quad (\text{S5})$$

when the same pool is used for both stages. The rates have different denominators and must not be differenced. Self-ranking prec.|recall on $n=500$ is 227/249 (91.2%), which is not forward-verify precision.

Forward-verify re-ranks by symmetric chamfer on ^{13}C peak sets $P, Q \subset \mathbb{R}$

$$d(P, Q) = \frac{1}{|P|} \sum_{p \in P} \min_{q \in Q} |p - q| + \frac{1}{|Q|} \sum_{q \in Q} \min_{p \in P} |q - p|, \quad (\text{S6})$$

where $\hat{P}(c)$ is predicted blind to the observed list P_{obs} . An instrumented diagnostic on a protocol slice is 58/65 (§8). The $n=500$ forward-verify wall is 204/45/251 (§8; data/fverify_n500/WALL_n500.md). The expand-500 official arm score is 103/230 verify vs 129/230 self (§8).

A pool estimator on an index set $\mathcal{S}$ is

$$\hat{p}_{\mathcal{S}} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} s(c_{i1}, y_i^*). \quad (\text{S7})$$

The ICLR headline uses $\mathcal{S}$ of size 500 (frozen qid list). The $n=519$ score_pooled.py print (243/519) is not used.

Confidence intervals on earlier validate-clean subsets are bootstrap 95% (score_main.boot, seed 0, 2000 resamples). Paired model differences use McNemar with Holm correction. Confidence intervals are not reported for $n=500$ or expand-500.

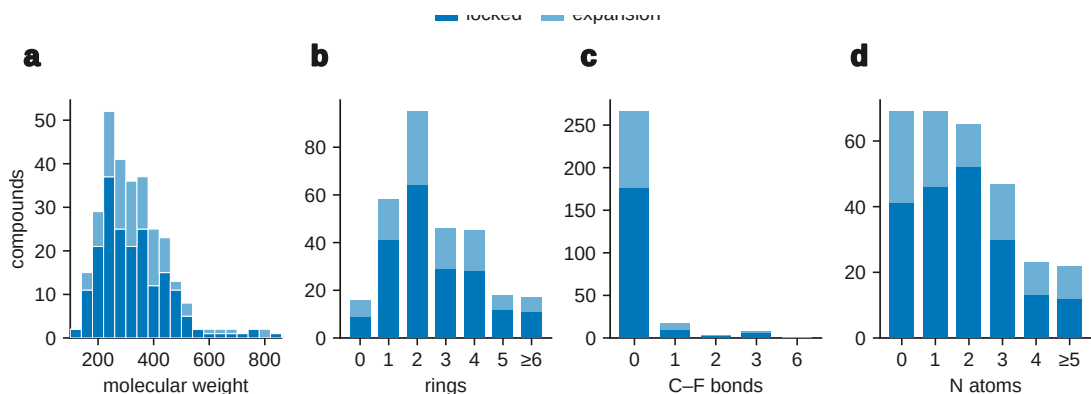


Figure S3: Molecular weight, ring count, C-F bonds, and nitrogen atoms for the $n=500$ literature cohort (median MW 306; median 2 rings). Moved from the main text to keep the ICLR body on budget.

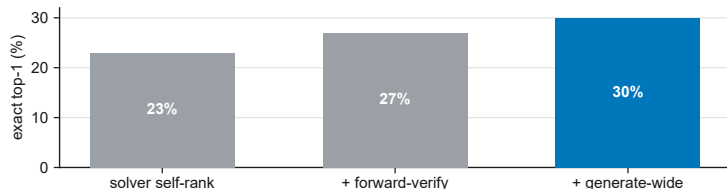


Figure S4: Inference ladder on the 60-compound arm: 23% $\rightarrow$ 27% $\rightarrow$ 30% (generate-wide recall 32% $\rightarrow$ 42%). Moved from the main text; integers unchanged.

5 Validate-clean extras

These rows are not the $n=500$ headline. Self-ranking precision | recall on $n=500$ is 227/249 (91.2%). ^{13}C -overread flags (R12, R22, R25, R82, R91) were marked before subset totals.

On an instrumented slice, of 137 analysable top-1 misses, 76.6% are constitutional isomers of the true structure; 22.6% share the true Murcko scaffold. Named-ring annotation on that slice is rare and does not drive the result.

6 expand-500 generation scores

Gold keys for this arm were used only for local scoring and are not in the review copy.

Mean best Tanimoto (score2 footer, all 230): simple 0.863; complex 0.645. Flagged qids (in 230, out of 224): R26, R31, R102, R105, R107, R138. Deposits: 230/230; predictions2.jsonl 230 lines, 690 candidates. Unique (formula, IR, ^{13}C) vs irexp_resolved.jsonl.gz: 230/230; 0 prior-round InChIKey-14 collisions. This arm used a thinking-tier consumer protocol and is not interchangeable with the no-thinking core protocol.

Verbatim score2 footer:

overall recovered (top-3): 138/230 (60%) top-1 exact: 129/230 (56%)
 simple : recovered 87/115 (76%) top1 82/115 (71%) meanBestTani 0.863
 complex : recovered 51/115 (44%) top1 47/115 (41%) meanBestTani 0.645

Table S2: Validate-clean extras show scaffold and stereo metrics under the same scorer, not the $n=500$ headline.

metric	overall	simple	complex
scaffold-level (best Tanimoto ≥ 0.45)	64%	80%	48%
mean best Tanimoto	0.66	0.79	0.53
full InChIKey (stereo) top-1	93/295 (31.5%) [26-37]	—	—
full InChIKey recall	108/295 (36.6%) [31-42]	—	—
corpus-reweighted top-1	26.5% [21-32]	—	—
corpus-reweighted recall	31.3% [25-37]	—	—

Table S3: Size bins show top-1 and recall on validate-clean $n=295$.

heavy atoms	n	top-1	recall
≤ 15	53	36/53 (67.9%)	39/53 (73.6%)
16-25	152	65/152 (42.8%)	75/152 (49.3%)
> 25	90	15/90 (16.7%)	16/90 (17.8%)

7 Headline generation scores

The estimator is Eq. S7. The paper headline is $n=500$. Roster: frozen qid lists in the code release.

Strata on the $n=500$ headline (same InChIKey-14 rule): simple 161/248 (65%) top-1 / 172/248 (69%) recall; complex 66/252 (26%) / 77/252 (31%). Self-rank precision | recall: 227/249 (91.2%). The $n=519$ pooled print (243/519) is not the headline. The $n=500$ forward-verify wall is 204/45/251 (§8).

8 Forward verification

8.1 $n=500$ wall

Classification: verified = recalled and chamfer-ranked first; misranked = recalled, distractor first; never-proposed = true structure not in the candidate set. The paper wall is the $n=500$ row. Arm rows below are not the headline.

Recalled = $204 + 45 = 249/500$, matching committed generation recall. Main-text Figure fig-wall reprints this 204/45/251 wall. Figure 1 reports the generation wall ($249/500$ recall $\times$ 227/249 self-rank = 45.4% top-1; wall 227/22/251).

8.2 Protocol-slice diagnostic

Source: data/diagnosis.json / scripts/forward_verify_all.py. This is a protocol-slice diagnostic, not a second $n=500$ wall.

Self-ranking 227/249 on $n=500$ is generation, not a substitute for 204/45/251.

8.3 Official expand-500 score

Blind ^{13}C campaign over 681 unique candidate SMILES (f1-f41 all present). Verify does not beat self-rank. Headline generation remains $n=500$ (227/500, 249/500). This arm score does not replace the $n=500$ forward-verify table.

Table S4: expand-500 scores show generation under score2 (InChIKey-14; confidence intervals not reported).

set	<i>n</i>	top-1	recall@3
all deposited Opus	230	129/230 (56.1%)	138/230 (60.0%)
simple	115	82/115 (71%)	87/115 (76%)
complex	115	47/115 (41%)	51/115 (44%)
validate-clean	224	127/224 (56.7%)	135/224 (60.3%)
simple	110	81/110 (74%)	85/110 (77%)
complex	114	46/114 (40%)	50/114 (44%)

Table S5: Headline scores show generation on the frozen *n*=500 roster (confidence intervals not reported).

set	<i>n</i>	top-1	recall@3
paper headline	500	227/500 (45.4%)	249/500 (49.8%)

Coverage: 41/41 batches; 681/681 unique SMILES; 230/230 qids fully covered. Multi-candidate only: all-230 self 128/137, verify 102/137; clean-224 self 126/134, verify 102/134. forward_verify_all.py was not run (no expand-500 diagnosis.json).

HOSE-style lookup on the protocol-slice candidate sets ties self-ranking; a modest GNN reaches 59/65 (91%) vs the LLM verifier’s 58/65 (89%). Deranging observed ¹³C pairings collapses precision (*p*=0.001).

9 Worked cases

The three locked-slice modes in the main appendix, plus additional panels that use already-reported numbers. Expansion qids reuse R22/R25. No SMILES are invented for R06/R26; only v3-R25 molecules are drawn (Figure S5).

9.1 Proposal miss: main-R06 / C₂₉H₃₀FNO₂

True acridinedione never enters the 3-candidate pool (Tanimoto-0.90 regioisomer: 4-fluorophenyl / phenyl swapped). IR 1641, 1510 cm⁻¹; ¹³C 195.7. Verifier unused. Re-ranking cannot recover a structure that was never proposed (Eq. S5).

9.2 Verify-save: v3-R25 / C₁₀H₁₄N₂O

True 2-pyridyl picolinamide at self-rank 3; self-rank 1 is 3-pyridyl. Forward-verify recovers (chamfer 0.42 vs 1.30 ppm, Eq. S6). IR 1679 cm⁻¹. Main-text Figure 3; SI Figure S5.

9.3 Verify-fail: v3-R26 / C₁₄H₁₁N₃O₃

True 3-nitrophenyl dihydroquinazolinone in pool (self-rank 2); 2-nitro ranked first. Chamfer 1.35 vs 1.36 ppm. IR 1652, 1530, 1352 cm⁻¹. Ranking can fail when predicted ¹³C lists are nearly tied (McNemar *p*=0.55; 7 gained, 4 lost on the protocol slice).

Table S6: The forward-verify wall shows the $n=500$ split (confidence intervals not reported; arm rows are not the headline).

arm	n	verified	misranked	never-proposed
locked protocol slice	194	58	7	129
Opus expansion	106	56	12	38
expand-500 cut	200	90	26	84
$n=500$ wall	500	204	45	251

Table S7: A protocol-slice diagnostic shows forward verification, not the $n=500$ wall (204/45/251).

	60-compound arm	protocol slice
generation recall	19/60 (32%)	65 (34%)
top-1, solver self-ranking	14/60 (23%)	55 (28%)
top-1, forward-verified re-ranking	16/60 (27%)	58 (30%)
precision recall — fverify	16/19 (84%)	58/65 (89%)
multi-candidate only — fverify	10/13 (77%)	30/37 (81%)
derangement floor on $n=37$	—	54.0%; fverify +27.1 points
McNemar vs self-rank	—	$p=0.55$ (7 gained, 4 lost)

9.4 Example payload

Main-text Listing 1 is the live IReXp JSON record ($C_{12}H_{11}BrN_4$; InChIKey ILJNJJNKEOAREX-UHFFFAOYSA-N; PubChem CID 57398578). SMILES, InChIKey, and licence fields are resource metadata. Bench solvers see only formula and printed peak lists. Gold constitution is scored only by Eq. S1. The listing is reprinted above with the same lstlisting+framed layout. Locked v3-R25 remains the mechanism worked case (main-text Figure 3).

9.5 Additional panels

Formula-only control ($n=60$). Formula only: 3/60 (5%) top-1 and 3/60 (5%) recall. Formula + IR + 1H + ^{13}C : 14/60 (23%) / 19/60 (32%). Outcomes perfectly nested (McNemar $p=0.001$).

Generate-wide (60-compound arm). Ten solver agents, up to six candidates each, pooled then re-ranked: recall 32% $\rightarrow$ 42%, top-1 23% $\rightarrow$ 30% (McNemar $p=0.34$); verification precision falls 84% $\rightarrow$ 72%.

IReXp-fine-tuned generator (locked slice). Pooled with Claude: recall 33.5% $\rightarrow$ 54.1%, top-1 28.4% $\rightarrow$ 35.1% (McNemar $p=0.015$). Training is the intervention that has been observed to move $P(\text{true} \in \text{pool})$.

Battery-electrolyte subset. $n=46$ literature electrolyte chemistry (not operando): 26% top-1, recall-bound.

Four-model Claude ladder ($n=24$). Haiku $\subset$ Sonnet $\subset$ Opus $\subset$ Fable is capability-sensitive but underpowered for adjacent ranks, with a protocol asymmetry.

Difficulty variation. Pre-registered draws differ in difficulty. The headline remains $n=500$ (Eq. S7).

Table S8: Official expand-500 chamfer scores show an arm result, not the $n=500$ wall and not the headline.

set	n	recall	self top-1	verify top-1	verify recall
all deposited	230	138/230 (60.0%)	129/230 (56.1%)	103/230 (44.8%)	103/138 (74.6%)
validate-clean	224	135/224 (60.3%)	127/224 (56.7%)	103/224 (46.0%)	103/135 (76.3%)

Table S9: The case grid shows three failure modes, with the same integers as the main-text case table.

mode	id / formula	notes
Proposal miss	main-R06 / $C_{29}H_{30}FNO_2$	True acridinedione never in the 3-candidate pool (Tanimoto-0.90 regioisomer: 4-fluorophenyl / phenyl swapped). IR 1641, 1510 cm^{-1} ; ^{13}C 195.7. Verifier unused.
Verify-save	v3-R25 / $C_{10}H_{14}N_2O$	True 2-pyridyl picolinamide at self-rank 3; self-rank 1 is 3-pyridyl. Forward-verify recovers (chamfer 0.42 vs 1.30 ppm). IR 1679 cm^{-1} . Figure S5.
Verify-fail	v3-R26 / $C_{14}H_{11}N_3O_3$	True 3-nitrophenyl dihydroquinazolinone in pool (self-rank 2); 2-nitro ranked first. Chamfer 1.35 vs 1.36 ppm. IR 1652, 1530, 1352 cm^{-1} .

10 Prompts and deposit schema

10.1 Headline prompt

Wrapping instruction text was not captured on the headline Claude arm (main-text Limitations (i)). Only the per-compound four-field payload is released. The skeleton below is the contract, not a recovered verbatim system prompt.

You are given only:
 formula (as from HRMS)
 ir_bands_cm-1 (author-transcribed experimental list)
 h_nmr, c_nmr (printed shift lists)
 Return ≤ 3 ranked SMILES.
 No name, no SMILES hint, no scaffold, no retrieved spectrum.
 Check molecular formula with RDKit before emitting a candidate.

Listing 1 payload (solver-facing fields only; gold held out):

```
{
  "formula": "C12H11BrN4",
  "ir_bands_cm-1": [3144.0, 3110.0, 3066.0, 2931.0,
    2863.0, 1632.0, 1589.0, 1565.0, 1485.0],
  "h_nmr": "8.83 (s, 1H), 8.04 (s, 1H), 7.81 (d, J = 8.8 Hz, 2H), ...",
  "c_nmr": "157.5, 140.3, 138.4, 132.4, 127.5, 120.4, 119.0, 116.4, 49.1"
}
```

10.2 Forward-verify prompt

For each candidate SMILES, predict a ^{13}C peak list without seeing the observed list. Re-rank by $d(\hat{P}(c), P_{obs})$ (Eq. S6). The instruction dispatched to the forward-prediction sub-agents was not captured; only the candidate SMILES and the resulting numeric lists are on disk for the protocol slice.

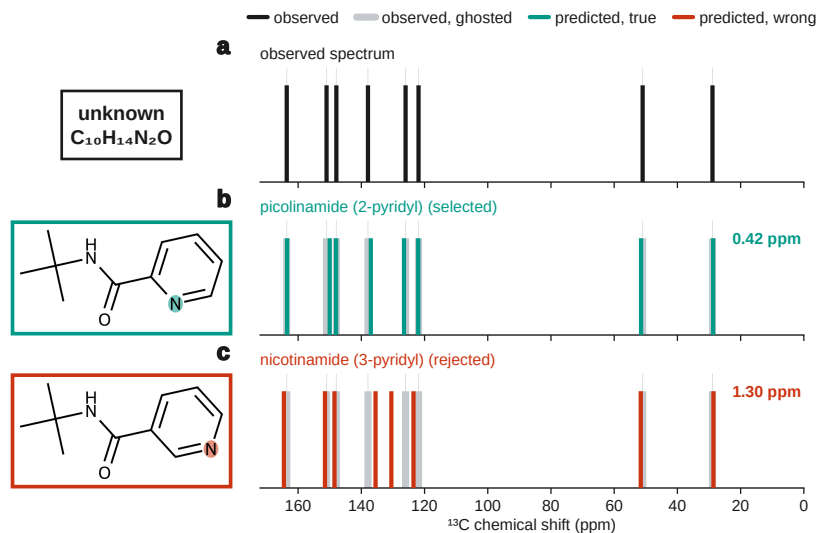


Figure S5: Case v3-R25 shows a verify-save (main-text Figure 3): self-rank prefers 3-pyridyl, while forward ^{13}C recovers 2-pyridyl (chamfer 0.42 vs 1.30 ppm).

10.3 Cross-vendor harness

Verbatim prompts in the code release (ESI S2) are the cross-vendor harness prompts, not the missing headline Claude wrapper. External submissions must document model version, prompt, and tool access (docs/LEADERBOARD.md).

10.4 Deposit schema

```
{"qid": "R01", "candidates": ["SMILES_rank1", "SMILES_rank2", "SMILES_rank3"]}
```

Score locally (code release):

```
python scripts/score_submission.py --predictions my_run.jsonl --name "YourModel-1.0"
python scripts/score_pooled.py --stereo # validate-clean stereo extras
python scripts/forward_verify_all.py # protocol-slice diagnostic
```

11 Model cards

11.1 Headline model card

11.2 Cross-vendor model card ($n=60$)

Verification precision exceeds generation recall in every arm. Bootstrap separates the paired gap for Claude (+52.5 points), GPT-5.6 Sol (+26.3) and Gemini (+23.3); Grok's gap (+9.2) is directional. A clean-clone control for Grok bounds key leakage (McNemar $p=0.39$). Reasoning-effort tier per vendor arm was not recorded.

11.3 Literature comparison

Any system reporting top-1 = a and top- k = b implies recall $\geq b$ and conditional precision $\leq a/b$. Script: scripts/literature_decomposition.py.

Table S10: The headline solver card shows recorded fields; unrecorded fields are left blank.

item	value
headline solver	Claude Opus, consumer subscription
thinking tier (core protocol)	none; a labelled thinking-tier arm exists
snapshot / seed / temperature / top_p / top_k / max_tokens	not exposed; not recorded
candidate budget	≤ 3 SMILES / qid
tool access	RDKit formula / parse check only; no web search
fine-tuning	none on the headline arm
paid API	no, for the core protocol
hardware	consumer harness; no cluster job for inference
scoring hardware	commodity CPU; RDKit InChIKey-14
cross-vendor arm	$n=60$; Grok 4.6, Gemini 3.7 Flash, GPT-5.6 Sol
Claude mean candidates on that arm	2.20 (others 3.00) — recall rankings approximate
expand-500 solver	Claude Opus thinking-tier (declared; not headline)
Incomplete additional-model deposits	not scored; not pooled
scoring scripts	scripts/score_pooled.py, score_main.py, bench-mark_v2.py score2

Table S11: Cross-vendor scores show generation recall versus verification precision (different denominators).

model	generation recall	prec. recall	multi-candidate only
Claude Opus	19/60 = 32%	16/19 = 84%	10/13 = 77%
Grok 4.6	32/60 = 53%	20/32 = 62%	20/32 = 62%
Gemini 3.7 Flash	30/60 = 50%	22/30 = 73%	22/30 = 73%
GPT-5.6 Sol	25/60 = 42%	17/25 = 68%	16/24 = 67%

Three groups changed only the data, and recall carried 68% (NMR-Solver simulated $\rightarrow$ real), 70% (SpecX random $\rightarrow$ scaffold) and 83% (Espejo education $\rightarrow$ industrial) of each collapse. The solver-alone precision column on $n=500$ is self-ranking (227/249); the forward-verify row is the instrumented protocol slice (58/65).

12 IRexp resource and anonymity

IRSpectra-Bench is the blind peak-list benchmark and the propose $\gg$ verify diagnosis. IRexp is the spectral peak-list resource those problems are drawn from: 121,233 records; 43,060 structure-linked; 33,201 full IR+ ^1H + ^{13}C +structure quadruples (irexp_resolved). File schema and aggregate counts ship with the review copy (SCHEMA.md, data/). Benchmark construction details beyond the main-text contract (spectral validation filters, prompt-leakage audit, electrolyte SMARTS) live in the code release.

Double-blind review copy: <https://anonymous.4open.science/r/peaklist-corpus-review-10C4/>. Named archival hosting is restored at camera-ready. Fine-tunes must use data/train_no_bench.jsonl.gz so benchmark InChIKeys are withheld.

Listing 1 is the IRexp-style JSON+molecule plate in the ICLR PDF.

Table S12: Literature rows show selected systems, connectivity-scored or as published. This is the home of the table moved off the ICLR body.

system (<i>n</i>)	top-1	top- <i>k</i> (<i>k</i>)	recall	prec.
this work, solver alone (500)	45.4%	49.8% (3)	49.8%	91.2%
this work + forward-verify (slice)	29.9%	33.5% (3)	33.5%	89.2%
NMR-Solver, literature (450)	52.9%	67.3% (10)	$\geq 67.3\%$	$\leq 78.6\%$
NMRAgent, literature (450)	61.6%	70.0% (10)	$\geq 70.0\%$	$\leq 88.0\%$
Espejo Morales, education (236)	80.9%	90.0% (5)	$\geq 90.0\%$	$\leq 89.9\%$
Espejo Morales, industrial (34)	20.6%	29.1% (5)	$\geq 29.1\%$	$\leq 70.9\%$
Alberts, 6–13 atoms, NIST IR (3,455)	63.8%	84.0% (10)	84.0%	75.9%
SpecX, scaffold split, simulated (99,439)	29.7%	50.6% (10)	50.6%	58.7%

13 Reproducibility

Frozen questions, predictions, and scorers ship with the code release. The $n=500$ generation totals 227/500 and 249/500 regenerate from the released deposits. The $n=500$ forward-verify table is §8. The protocol-slice diagnostic is regenerated by python scripts/forward_verify_all.py.

Expand-500 gold keys are withheld from the review copy. The official expand-500 chamfer score is 103/230 verify vs 129/230 self.

The IRexp bibliographic entry is anonymous in the review copy; the dataset URL in the PDF is the anonymous.4open review copy. Fine-tunes should use data/train_no_bench.jsonl.gz.

Inference is not bit-exact: no pinned snapshot, temperature, or seed was recorded. Distributional agreement of frozen deposits is the intended claim. Unrecorded fields: headline wrapper text; model snapshots; decoding parameters; sampler seeds.

On-bench scores for SpectraLLM, IR-Agent, Spectro, NMIRacle, Alberts, and CASE were not obtained; an expert-chemist audit and a leave-one-modality arm were not run.